

Master Thesis — Progress Report

Spain's MTU15 reform: empirical assessment in progress

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Outline

- Objective and motivation
- Market structure and reform context
- Research questions: *a priori* ideas
- Data pipeline
- Research questions: what worked and what did not
- Next steps: what is left and what is not feasible

Objective and motivation

- The purpose of my thesis is to empirically assess the impact of Spain's 2024–2025 **market time unit (MTU15) reform sequence in the electricity wholesale markets**, focusing on firm strategic behaviour and market outcomes and building a theoretical framework to better capture the mechanism.
- Due to the high institutional and data complexity of the reform, I have focused on building a robust data pipeline, as well as understanding the market structure and reform context in depth, before attempting to test any specific hypothesis.
- Moreover, we have to be very careful with how we read the data, since the reform sequence is not cleanly separated from other confounding events (gas-price normalisation, blackout and post-blackout regime, etc.), so **institutional awareness** is key here.

Market structure

HERE A COMPLEX GRAPH OF THE MARKETS, INCLUDING OMIE (DA, IDA, CONTINUOUS), REE RESERVE MARKETS, IMBALANCE SETTLEMENT. ALSO MENTION THE INTERVENTION OF THE TSO IN THE OMIE PROGRAMS!!

Reform context

- Traditionally MTU was **60 min** everywhere.
- Three stages:
 - **Dec 2024** — **ISP15**: imbalance settlement → 15-min.
 - **Mar 2025** — **ID15**: intraday → 15-min.
 - **Oct 2025** — **DA15**: day-ahead → 15-min.
- Two transitional regimes where *settlement* and *trading* clocks differ.
- Plus: post-2025-04-28 blackout ⇒ *operación reforzada* (REE's reinforced-operation regime), continuous since.

Period	ISP	ID	DA
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	15	60
Oct25 onward	15	15	15

ISP15

Dec 2024

ID15

Mar 2025

DA15

Oct 2025

Research questions: *a priori* ideas

Data pipeline: what I have and why (I)

Data pipeline: what I have and why (II)

Research questions: what worked and what did not

- BIDS RESULTS: CCGT EXPLOIT MTU.

Next steps: what is left and what is not feasible

Thank you!

Slide 12 — Identification: within-day critical-vs-flat hours DiD

The core design idea. The MTU15 reform doesn't change market rules uniformly across hours of the day — it changes them *differentially*. Granularity reform (60-min $\rightarrow$ 15-min) only bites economically in hours where the within-hour residual-demand profile is steep enough that 15-min slicing creates strategic optionality. In flat hours, 15-min and 60-min are mechanically equivalent.

Within-day treatment-control structure:

- **Treated unit:** critical hours $h \in \{7, 8, 16, 17, 18\}$. Top 5 by within-hour residual-demand variance (σ_{within}^2). Steep ramp hours where granularity should bite.
- **Control unit:** flat hours $h \in \{3, 4, 5\}$. Pre-dawn hours, bottom of σ_{within}^2 ranking. Genuinely flat within-hour profile, no strategic optionality from granularity.
- **Treatment timing:** reform regime indicator (DA15/ID15 vs pre-IDA).
- **Estimand:** the causal effect of the reform on the granularity mechanism, identified by how the critical-flat differential changes from pre-IDA to DA15/ID15.

Specification:

$$q_{ift} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_t + \beta_3 (\text{crit}_h \times \text{post}_t) + \gamma_f + \delta_d + \varepsilon_{ift}$$

Big-4 firm f , day t (Day-of-Week FE δ_d), hour h . Cluster-robust SE by date.

Slide 13 — Five identification advantages over standard reform DiD

1. **Within-day FE absorbs all daily-level shocks.** Critical and flat hours of the same day share the same Spanish wind regime, the same temperature, the same hydro inflow, the same fuel prices, the same regulatory announcements. Anything operating at the daily level differences out by construction.
2. **Calendar mix controls itself.** Critical and flat hours come from the same dates. Identical calendar mix for treated and control by construction.
3. **Weather differences out at the hour-pair level.** Same-day hours share most weather conditions.
4. **Built-in parallel-trends check.** The pre-IDA crit-flat differential is the placebo test: if pre-existing crit-flat structural differences are large and unstable, the design is in trouble.
5. **Treatment assignment is mechanism-targeted.** The contrast is “hours where granularity should matter” vs “hours where it shouldn’t” — not “post-reform vs pre-reform”. Identifies the granularity-mechanism causal effect specifically.

What within-day FE does NOT solve: within-hour-of-day trends across calendar time. Spanish solar grew $\sim 6\times$ since 2018; renewable share at $h\{7,8,16,17,18\}$ vs $h\{3,4,5\}$ has a calendar-time trend. Conditional parallel trends would require absorbing this. Tonight’s test: it does *not* hold even with $\text{wind} + \text{solar} + \text{demand} \times \text{hour}$ controls.

Slide 14 — Headline empirical result: q_2 V-shape

Outcome variable: q_2 = signed Big-4 IDA repositioning per firm-day-hour, MWh. From PIBCIE, summing IDA-cleared assigned_power_mw $\times$ mtu_minutes/60 across periods within each (date, firm, hour). Positive q_2 = firm sells more in IDA above DA commitment; negative = firm buys back.

Spec: $q_2 \sim \text{crit} + \text{post} + \text{crit} \times \text{post} + \text{firm FE} + \text{DOW FE}$, cluster by date. Treated $h \in \{7, 8, 16, 17, 18\}$, control $h \in \{3, 4, 5\}$. Pre-IDA = before 2024-06-14. Post = DA15/ID15 (Oct 2025+).

Result with clean flat-hour controls:

Coefficient	Estimate	SE	p-value
crit_pre (pre-IDA crit-flat gap)	+41.21	3.36	1.6×10^{-34}
DiD δ (crit $\times$ post)	+58.67	12.18	1.5×10^{-6}
crit_post (DA15/ID15 crit-flat gap)	+99.80	—	—

Interpretation. At DA15/ID15, Big-4 firms reposit **+100 MWh more** per firm-hour in IDA in critical hours than in flat (pre-dawn) hours. The reform-attributable component (DiD δ) is +59 MWh.

Sample: 78,177 firm-day-hours. Two firms (GN, IB) carry most of the result; see slide 22.

Slide 15 — Same-calendar-month robustness

The concern this addresses. Pre-IDA spans 2018–2024 (all calendar months). DA15/ID15 spans Oct 2025+ (only Oct–May months). Calendar mix differs $\Rightarrow$ Spanish seasonality could bias the comparison.

The fix. Restrict pre-IDA observations to the same calendar months as DA15/ID15 (Oct–May only). Re-estimate the same DiD spec.

Outcome: q_2 as above (Big-4 IDA repositioning per firm-hour, MWh). Note: this run uses the original definition with all-non-critical-hours as control (the +83 baseline rather than +100 with clean controls).

Pre-IDA window	N_pre	DiD δ	SE	p-value	ratio vs full
Full (default)	224,061	+79.36	6.71	3×10^{-32}	100%
Same-cal-month	—	+76.5	6.77	1×10^{-29}	96%

Verdict. The headline coefficient survives same-calendar-month restriction at 96% magnitude. Calendar mix is not driving the result.

CLAUDE.md verdict criterion (“>50% magnitude with same sign”) easily met.

Slide 16 — Baseline-window sensitivity (pre-trend stability)

The concern this addresses. The 6.5-year pre-IDA window averages over many distinct conditions (pre-COVID, COVID, energy crisis, normal). If the pre-IDA crit-flat gap drifted, the DiD identification is fragile.

Test. Re-estimate the DiD with progressively narrower pre-IDA windows. With clean flat controls $h\{3, 4, 5\}$:

Pre-window	months	N_pre	crit_pre (SE)	DiD δ (SE)	crit_post
full	78.5	74,700	+41.2 (3.4)	+58.6 (12.2)	+99.8
excl_crisis	54.2	51,414	+72.9 (4.2)	+27.0 (12.4)	+99.9
recent_2y	24.4	23,273	-33.4 (4.4)	+132.6 (12.5)	+99.2
recent_1y	12.2	11,625	-17.1 (5.7)	+116.4 (13.0)	+99.3
recent_6m	6.1	5,778	-13.8 (8.2)	+112.8 (14.3)	+99.0

Two readings:

- **Within-DA15/ID15 fact (crit_post = +99.8) is rock-solid.** Constant across all five baseline choices — this is what the design identifies.
- **Pre-trend is real and not stable.** crit_pre drifts from +73 (deep history) to -33 (recent_2y) — 106 MWh swing. DiD δ ranges +27 to +133 across windows. Reform-attributable component is a range, not a point estimate.

Slide 17 — Per-hour DiD reveals two-cluster mechanism

Per-hour DiD with control = $h\{3, 4, 5\}$, **outcome** = q_2 . For each hour h , treat that single hour vs the flat-hour control set. Reveals which hours actually carry the reform signal:

Cluster	hours	DiD δ range	p-values	sign
Morning ramp + peak	h7, h8	+64 to +68	$< 10^{-6}$	+
Evening peak	h17, h18, h19	+68 to +101	$< 10^{-7}$	+
Midday plateau	h10–h15	−48 to −131	$< 10^{-7}$	−
Late-night transition	h0–h2	+25 to +46	$< 10^{-4}$	+ (weak)
Other	h6, h9, h16, h20–h22	near zero	ns	—
h23 (late edge)	h23	−70	5×10^{-6}	−

Two-cluster mechanism:

- **Ramp + evening-peak hours: positive DiD δ .** Big-4 reposit MORE in IDA under DA15/ID15. Consistent with Ito–Reguant: residual demand rising sharply within the hour creates strategic withholding amplified by 15-min granularity.
- **Midday plateau (h10–h15): negative DiD δ , large magnitude.** Big-4 reposit LESS (or buy back) in IDA. Solar abundance crowds out conventional commitment; 15-min granularity allows precise quarter-by-quarter reduction in commitment.

Implication. The mechanism is bidirectional — granularity amplifies critical-hour repositioning, in both signs depending on whether residual demand is rising (scarcity) or falling (abundance) within the hour.

Slide 18 — Cumulative critical-set sensitivity

Why this matters. The canonical 5-hour critical set comes from a top-5 cut of σ_{within}^2 . Why stop at 5? The σ_{within}^2 ranking after rank 5 is $h\{12, 6, 9, 11, 13, 14, 10\}$. Adding any of those should clarify if 5 is the right cut.

Cumulative DiD with control = $h\{3, 4, 5\}$, outcome = q_2 :

Top K critical hours	added	DiD δ (SE)	crit_post	p-value
K=1: {17}	—	+67.7 (16.7)	+113.5	5×10^{-5}
K=4: {7,16,17,18}	ramp + evening	+57.4 (12.7)	+100.5	6×10^{-6}
K=5: canonical {7,8,16,17,18}	+ h8 (morning)	+58.6 (12.2)	+99.8	2×10^{-6}
K=6: + h12	adds midday	+30.5 (11.6)	+71.1	9×10^{-3}
K=11: + $h\{6,9,11-14\}$	full midday + ramp neighbours	-11.4 (10.2)	+26.7	ns
K=12: + h10	also midday	-14.5 (10.1)	+23.5	ns

Verdict. The +100 result is robust to dropping any single canonical hour, and to cutting at K=4 (without h8) or K=1 (just h17). **Adding h12 (rank 6) collapses the coefficient by 30%**, because h12 is in the opposite-sign midday cluster. The σ_{within}^2 ranking is a necessary but not sufficient hour selector — it doesn't distinguish “hours where granularity creates rising-demand scarcity” from “hours where granularity creates falling-demand abundance”. The canonical 5-hour set happens to align cleanly with the rising-demand cluster.

Slide 19 — Conditional parallel trends test

The test. If the pre-trend in `crit_pre` (slide 16) is driven by Spanish solar capacity growth, demand growth, or wind variability $\times$ hour-of-day, then conditioning on these should stabilise the DiD δ across baseline windows.

Specs run:

- A: sparse (firm + DOW FE only)
- B: + wind_GWh + solar_GWh + each $\times$ critical (interactions with treated dummy)
- C: + demand_GWh + demand $\times$ critical (additional)

Wind/solar from ENTSO-E A75; demand from ENTSO-E load_actual. UTC $\rightarrow$ Madrid local time, aggregated to hour.

DiD δ across baseline windows, by spec:

Spec	full	excl_crisis	recent_2y	recent_1y	recent_6m	range
A: sparse	+58.7	+27.1	+132.6	+116.4	+112.8	106
B: + wind, solar ($\times$ crit)	+50.4	+28.7	+121.5	+107.2	+107.4	92
C: + ren + demand ($\times$ crit)	+40.1	+10.6	+123.6	+97.2	+101.2	113

Verdict: conditional parallel trends does NOT hold. Renewable controls absorb only $\sim 13\%$ of the pre-trend instability (range 106 $\rightarrow$ 92). Demand controls inflate SEs heavily without further stabilising point estimates. The pre-trend is real and is driven by something my candidate controls don't capture — candidates: CCGT/coal retirement, Rule 28.8 anticipation, structural shifts in dispatchable supply.

Slide 20 — Per-firm × tech decomposition

Outcome: q_2 apportioned to (firm, tech) via DA-sell-share weighting. crit-flat differential at DA15/ID15 in MWh per firm-hour:

	CCGT	Hydro	Nuclear	Firm total
GN	+163.5	+149.1	—	+312.6
IB	+2.5	+8.6	+84.1	+95.2
GE	−0.7	+5.2	+6.0	+10.5
HC	−24.2	+9.2	—	−15.0

Reading.

- **Naturgy carries the result.** GN's CCGT and Hydro both engage strongly in critical-hour IDA repositioning. GN total +313 dwarfs every other firm.
- **Iberdrola contributes through Nuclear.** IB's CCGT (+2.5) and Hydro (+8.6) are essentially zero. The +95 IB total is +84 from the nuclear-apportioned slice. **Caveat:** this is firm-level q_2 multiplied by IB's DA nuclear sell-share, NOT direct evidence of nuclear-plant-level repositioning. The apportionment is mechanically large for nuclear because nuclear is large in IB's DA portfolio.
- **GE near-zero (+10.5); HC slightly negative (−15).** Functionally placebos: small market share, no IR-amplification response.

Implication for narrative. The granularity-amplifies-IR mechanism appears strongly in one firm (GN) and weakly through one tech in another (IB Nuclear). Cross-firm response is heterogeneous. “Big-4 strategic IDA repositioning” overstates uniform engagement; the data say “Naturgy responds; IB partially through nuclear; GE/HC do not”.

Slide 21 — CCGT bid-shape: rich-ladder pattern post-MTU15-DA

Independent of the supply-quantity story. Post-MTU15-DA (Oct 2025+), Big-4 CCGT bid LADDERS differ qualitatively across the 4 quarters of an hour — and the difference concentrates in critical hours.

Per-quarter bid measures (CCGT, qty-weighted), critical vs flat:

Measure	flat	critical	direction
Number of price tranches per quarter	5.08	5.50	+8% in critical
Ladder slope $(p_{\text{top}} - p_{\text{bot}})/q_{\text{tot}}$	1.22	1.38	+13% steeper
qty-weighted avg price (€/MWh)	488	472	LOWER in critical

Distributional view. $CV(p_{\text{avg}})$ across 4 quarters of an hour:

Tech	P90 flat	P90 critical	ratio
CCGT	1.6%	7.9%	4.9×
Hydro	1.6%	3.6%	2.3×
Wind	3.1%	3.9%	1.2× (essentially flat)
Solar	0.0%	0.3%	— (very low absolute)
Nuclear	2.4%	2.6%	1.1× (essentially flat)

Interpretation. Pre-MTU15-DA, the firm could only set one price for the hour. Post-MTU15-DA, the firm uses the granularity to bid a richer ladder in critical hours: more low-price inframarginal tranches at the bottom (push avg down, lock clearing) PLUS a steeper high-price tail (capture scarcity rent). **Wind/solar/nuclear do not engage** despite Ito–Reguant downward-curtailement optionality — the strategic story is concentrated in dispatchable thermal/hydro where ramping costs make 15-min granularity economically meaningful.

Slide 22 — CCGT extensive-margin composition shift (DA60/ID15-prebo)

Number of unique active Big-4 units in critical vs flat hours, by regime (PDBCE auction-cleared):

Regime	flat units	critical units	gap (crit–flat)
pre-IDA	95	95	0
3-sess	72	68	–4
ISP15-win	69	67	–2
DA60/ID15-prebo	53	40	–13
DA60/ID15-reforz	69	66	–3
DA15/ID15	68	66	–2

The 13-unit gap under DA60/ID15-prebo is entirely CCGT. Hydro / nuclear / thermal-nonRE all match across critical and flat (zero or near-zero gap by tech). Decomposed:

Tech (gap crit–flat)	3-sess	ISP15-win	DA60-prebo	DA60-reforz	DA15/ID15
CCGT	–2	–1	–13	–2	–2
Hydro / Nuclear / Other	0	0	0	0–1	0

Reading. 13 Big-4 CCGT units stopped showing up in critical-hour DA auctions during the asymmetric-clock window (Mar 19–Apr 27 2025). Units returned under *operación reforzada* (post-blackout): REE compelled CCGT back into critical hours via PO-3.x voltage-stability requirements. Under symmetric-clock DA15/ID15, units stay in critical hours voluntarily.

Consistent with the asymmetric-clock-induced hedging-cost story: CCGT operators face the worst within-hour mismatch cost (steepest ramps, highest fuel-cost-of-deviation); a portion exit critical hours when the friction is binding.

Slide 23 — Honest synthesis: what's robust, what's not, what's next

What's robust (identification-clean):

- **Within-DA15/ID15 critical-flat q_2 gap = +100 MWh per firm-hour**, control = $h\{3, 4, 5\}$. Constant across all baseline windows, control sets, conditional specs.
- **Two-cluster mechanism**: ramp + evening peak hours (positive δ); midday plateau (negative δ). Both directions of granularity-mediated repositioning.
- **CCGT bid-shape rich-ladder pattern** post-MTU15-DA. Independent of supply-quantity story; survives on its own. Wind/solar/nuclear do not engage — mechanism is dispatchable-thermal-specific.
- **CCGT extensive-margin composition shift** under DA60/ID15-prebo (13 inactive Big-4 units in critical hours; gap closes under reforzada).

What's not robust (caveats to acknowledge):

- **Reform-attributable component (DiD δ) is a range, not a point estimate**. +27 to +133 across baseline windows. Pre-trend in crit_pre is real (+73 to -33 across pre-IDA sub-windows).
- **Conditional parallel trends with wind+solar+demand $\times$ hour-of-day does NOT hold**. Renewable controls absorb only $\sim 13\%$ of the pre-trend.
- **The +100 is concentrated in Naturgy**. GE/HC are essentially placebo; IB contributes only through nuclear apportionment artifact.

Next steps for the thesis (not for tomorrow):

- Construct CCGT/coal capacity series and re-run conditional spec.
- Regression discontinuity around Oct 1 2025 (sharp MTU15-DA boundary) — avoids the parallel-trends problem.
- Event study with explicit pre-trend modelling.